Using Convolutional Neural Networks for Fingerspelling Sign Recognition in Brazilian Sign Language

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ABSTRACT

Deaf people communicate naturally using sign languages, and because of this, they have difficulty in communicating using oral or written languages. To minimize this problem, one alternative is to use machine translation systems from sign languages into oral languages, especially for scenarios where human interpreters are not viable or unavailable. In this work, we address this problem by proposing a solution for fingerspelling recognition in Brazilian Sign Language (Libras) using Convolutional Neural Networks. The system uses a 224000 images dataset created by our team, which represents the letters of the Libras alphabet signed by 12 people in different backgrounds, body arm, hand positions, and lighting patterns. The results show that the solution had an average accuracy of approximately 99% in a dependent person scenario, and had an average accuracy of 71% in an independent person scenario. This type of solution can be used together with machine translators from Brazilian Portuguese to Libras, to assist in the communication of Brazilian deaf people.

CCS CONCEPTS

• **Human-centered computing** → **Accessibility design and evaluation methods**; *Accessibility systems and tools*; • **Applied computing** → **Language translation**; • **Social and professional topics** → *People with disabilities.*

KEYWORDS

Deep Learning, Hearing impairment, Accessibility, LIBRAS, Keras, FFmpeg, Sign Language Recognition, CNN

ACM Reference Format:
Douglas F. L. Lima, Armando S. Salvador Neto, Ewerton N. Santos, Tiago Maritan U. Araujo, and Thais Gaudencio do Rêgo. 2019. Using Convolutional Neural Networks for Fingerspelling Sign Recognition in Brazilian Sign Language. In *Brazilian Symposium on Multimedia and the Web (WebMedia '19), October 29-November 1, 2019, Rio de Janeiro , Brazil.* ACM, New York, NY, USA, 7 pages. https://doi.org/10.1145/3323503.3360306

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WebMedia '19, October 29-November 1, 2019, Rio de Janeiro , Brazil
© 2019 Association for Computing Machinery.
ACM ISBN 978-1-4503-6763-9/19/10…$15.00
https://doi.org/10.1145/3323503.3360306

1 INTRODUCTION

Hearing impairment affects a large number of people in the world. According to the World Health Organization (WHO), in 2018, about 466 million people worldwide have hearing impairment, and 7% of them (approximately 34 million) are children [4]. It also estimates that the number of people with a hearing impairment may increase considerably over the years, possibly reaching 900 million in 2050 [4].

In Brazil, a significant portion of the population also has a hearing impairment. According to the Brazilian Institute of Geography and Statistics (IBGE), in 2010, approximately 9.7 million people had some hearing impairment, and 22% of them had a severe degree of disability [9].

Many deaf people have difficulties in understanding and communicating through texts in written languages. Since these languages are based on sounds, most of them spend several years in school and fail to read and write in the spoken language of their own country [20]. Reading comprehension tests performed by Wauters [21] with deaf children aged 7–20 in the Netherlands showed that only 25% of them read at or above the level of a nine-year-old hearing child.

Thus, deaf people communicate naturally through spatial-visual languages called sign languages, and the spoken languages used by the majority of the population are just a "second language". As a result, to reduce the barriers to communication and access to information for deaf people, a process of translation/interpretation between spoken and sign languages (SLs) is usually necessary.

SLs are natural languages, with their lexicon and grammar, developed by deaf communities over time. Thus, there is no unique SL. Usually, each country has its own sign language, some even more than one. By 2013, there were already over 144 sign languages cataloged around the world [22].

In the scientific literature, there are some works addressing the difficulties of communication and access to information of the deaf people. Some works involves solutions for translating spoken language content into sign languages (eg. Brazilian Portuguese to Brazilian Sign Language - Libras) [11], whereas others involve recognizing/ translating sign language content into spoken languages

(eg, Libras to Brazilian Portuguese), the latter that is the focus problem of this paper.

Munib et al. [12], for example, used the Hough transform and neural networks to recognize ASL signs from images, without gloves. The authors' proposal is to contrast edges and smooth the change of melanin from hand to hand, allowing the user to interact with the system more naturally.

There are also solutions focused on ASL translation using continuous video models. Ye et al. [10], for example, used a hybrid model that aggregates 3D Convolutional Neural Networks (3DCNN) and Recurrent Neural Networks (RNN) to address this problem. In their solution, the 3DCNN captures RGB image characteristics, motion and depth channels, and the RNN captures temporal information from small videos.

These works, especially the related with recognition from sign languages content generally have some restrictions, such as the dependence of gloves, armbands or visual marking on images; the use of dataset images with a specific background and predefined color patterns. Besides, we did not find any work involving the recognition of signs for the Brazilian Sign Language (Libras).

Thus, to address these issues, in this work, we propose a solution for fingerspelling recognition in Libras using Convolutional Neural Networks. To develop the solution, we create a dataset containing approximately 224000 images, which represents 26 letters of the Libras alphabet signed by 13 people in different backgrounds, body arm and hand positions, and different lighting patterns. Another important point is that our solution does not use any gloves, armbands or visual markers in the images, i.e. it recognizes the Libras gestures using just the images. The results shows that the solution had an average accuracy of approximately 99% for the best CNN model in a dependent person scenario and had an average accuracy of 71% in an independent person scenario.

This type of solution can be integrated with machine translators of text and audio from Brazilian Portuguese to Libras (eg. VLibras, Handtalk or Rybená, etc.), to assist in the communication of Brazilian deaf people.

It is also important to note that the proposed solution is not intended to replace human interpreters. The proposal is to have a practical solution that can assist both deaf and normal people to reduce barriers to communication and access to information, especially in settings where human interpreters are not viable or unavailable.

In fact, we know that fingerspelling is not the ideal setting for communication with deaf people. However, given the current difficulties in communicating with Brazilian deaf people, and the complexity of developing a solution for recognizing thousands of Libras signs, expressions and/or sentences, we initially chose to investigate the fingerspelling recognition of Libras as a first solution for this kind of problem. One of our future work proposals is to evolve this database and the solution so that it can also be applied to recognize signs, expressions and sentences in Libras.

The remainder of the paper is organized as follows. Section 2 presents some works related to machine translation from sign languages into oral languages. Section 3 presents the proposed solution, and the methodology used to develop it. In Section 4, we present some tests to evaluate the proposed solution. Final remarks are presented in Section 5.

2 RELATED WORKS

Usually, we have three approaches for sign language recognition in the scientific literature. The first one is based on sensors, where the classification of the signs is based the several information generally collected from sensors, such as gloves or armbands. Some examples of information collected by these sensors are handshapes, hand orientation, hand movement, and facial expressions, among others [1, 2]. This type of solution is sensor-dependent, which can make it difficult to use the solution in some real practical situations, especially when the sensor is not available.

The second approach is generally based on computer vision, where researchers and developers try to classify or recognize the signs using just hand image. As mentioned in Section 01, Munib et al. [12], for example, use the Hough Transform, a mathematical technique that performs detection of geometric shapes in digital images, and Neural Networks for fingerspelling recognition in American Sign Language (ASL), without glovers, armbands or visual marking. They used a database composed of 15 images of each letter of the alphabet in ASL (fingerspelling) and had an accuracy of 92.3%. However, their dataset images contained only a constant black background, which also makes it difficult to apply it in a real practice situation.

Ye et al. [10] presented a solution for recognizing ASL gestures in a continuous video. They used a dataset consisting of 27 videos from 15 different people for 99 individual ASL signals, representing words rather than letters, and obtained a 69.2% accuracy in recognition.

The third approach combines the use of sensors (eg., gloves) and cameras. Bhatnagar et al. [3] proposed to merge previous approaches using hardware and software for translating sign language into speech, using a base with videos and motion data captured by the gloves. They tested with 2250 examples, and obtained an accuracy of 71.3%. They also performed a test excluding similar signs from the dataset, increasing their accuracy to 78.2%. However, there are few works using this approach, due to the cost and computational overhead of the entire configuration.

3 PROPOSED METHOD

As mentioned in Section 1, in this work, we propose a solution for fingerspelling recognition in Brazilian Sign Language (Libras) using Convolutional Neural Networks (CNNs). Our solution does not use any gloves, armbands or visual markers in the images, recognizing these Libras gestures using just images, and considering different backgrounds, signers, hand positions, lighting patterns, among others, trying to approach real practical situations for Brazilian deaf users.

In Section 3.1 and 3.2, we will present the dataset[23] created to develop the proposed solution, and the CNN models we used. Section 3.3 describes how we conduct the training and testing of these models.

3.1 Creation of the Dataset

Initially, we create a dataset of videos that aims to get closer to real practical situations, considering different environments, backgrounds, signers, body, arms and hand positions, lighting patterns, among others. The videos were recorded using three smartphones

from different manufacturers and camera qualities. The smartphones used were as follows:

- LG X Style: Resolution of 3264 x 2448 pixel with a total of 8 megapixels;
- Quantum Fly: Resolution of 4608 x 3456 pixel with a total of 16 megapixels;
- Moto Maxx XT1225: Resolution of 5248 x 3936 pixel with a total of 21 megapixels.

It is important to point out that we do not found any similar datasets (with this type of variation) for Libras in the scientific literature. To build the database, initially we recorded videos for each of the 26 letters of the alphabet in Libras with 13 users of different age, gender and ethnicity, which results in 338 recorded videos.

The method used to capture the 26 Libras letters (or signs) was to start with the hand centered on the camera and, while the video was recorded, the participants were guided to perform variations in hand position, scale (i.e.,vary the distance from hand to camera), and also perform some hand rotations. Each video had a duration of about 25 seconds.

Afterwards, we preprocess the videos with the FFmpeg [19] software, extracting 20 images per second of the videos collected. Some examples of these images are shown in Figure 1.



Figure 1: Some examples of each letter from the developed dataset.

Afterwards, we apply a data augmentation in these images/samples, using Keras 2.2.4 [17]. The parameters applied in this data augmentation were as follows:

- Zoom factor of 0.2 at maximum.
- Increase/Decrease of illumination with a range maximum of 30% of the original illumination.
- Shear factor of 0.8 at maximum.
- Rotation with a range of 0 to a maximum of 10° in clockwise or counterclockwise of the original image.
- Mirroring the samples.

Some results of these operations can be shown in Figure 2.

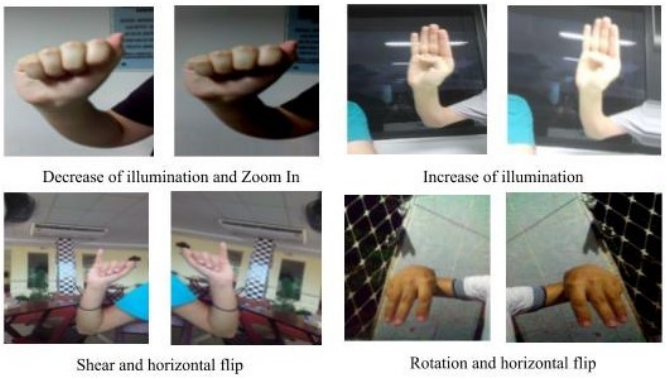


Figure 2: Some examples of images of the dataset after applying data augmentation.

The resulting final dataset consists of approximately 224000 images, from the 26 letters of the Libras alphabet signed by 13 persons in different backgrounds, body arm, and hand positions, and different lighting patterns were generated. The dataset is available in[23].

3.2 CNN Models

Since our final goal is a solution that can be used by Brazilian deaf people in real situations and ubiquitously, we focus on the use of neural network architectures that were efficient for use in smartphones or embedded systems, but which also presented good results in competitions or in works in other areas of computer vision [16], such as the MobileNet [5], with 71% in ImageNet[18] and NASNetMobile [6], which was proposed more recently and has better results in ImageNet [18] (74.1%) when compared to MobileNet. However, for comparison, we also perform tests on more robust architectures like VGG16 [7], InceptionResNetV2 [8], respectively with accuracy in ImageNet [18] of 71% and 80% in NasNetLarge [6] which is also more recent than the other two cited and achieved state of the art accuracy of 83% in the same dataset.

To optimize training time, we chose to use pre-trained models (using the ImageNet database [18]) of these networks. Thus, the weights of the pre-trained models were used as initial values in our training. Only the weights of the input layer were randomly initialized since we reduced the size of the images before training to 160x160, which was do empirically the lowest resolution obtained in our tests, which not compromise the recognition of gestures at the end of the training. However, it is important to note that all layers of the network were re-trained, and the values of the pre-trained models were used only as initialization values of our model.

3.3 Training and Validating the Models

To verify the accuracy of the models and check whether they can generalize different people, we follow the method defined by Ye et al. [10], where the training and validation were performed in two different ways:

- Dependent Person: The image of the hand of some people who are in the training set were randomly selected to compose the validation set. This was done to know the learning ability that each architecture had of the data set created and the batch method we used.
- Independent Person: The image of the hand of some people who were also randomly selected to compose the validation base and were excluded from the training set. This separation was made to understand how each model was generalized in a situation closer to a real case, with the image of a person's hand in a scenario that was not present in their training.

The training and tests were performed using a computer with an Intel(R) Core(TM) I3-3240 CPU @ 3.40GHz processor, 32 GB of RAM DDR3 and a Quadro P6000 24Gb Gddr5X 384 Bits graphic card.

Before training, we also use a stratified collection of samples in batches, due to the difficulty in in managing the use of memory in the platform that we use for training and testing. More specifically, for each of the 26 letters of the alphabet, the largest possible number of images were taken, which the memory of the training machine would be able to support, leaving only free space for the processing of main functions of the operating system, and other operations.

For example, for the VGG16 model, we were only able to run 44,800 instances (images) per batch. Thus, we had to execute five batches of 44,800 images to complete one eppoch (224 thousand images).

4 RESULTS AND DISCUSSION

To validate the proposed solution, we performed a set of tests using the created dataset. Initially, we performed a test to calculate the average accuracy for fingerspelling recognition in Libras of the different CNN models, considering the Dependent and Independent Person scenarios. The results of this test are presented in Table 1.

Model	Dependent	Independent
MobileNet[5]	**85.23%**	**61.35%**
NASNetMobile[6]	79.26%	39.86%
VGG16[7]	**99.73%**	**70.21%**
InceptionResNetV2[8]	93.32%	48.65%

Table 1: Average accuracy for Dependent and Independent Person scenarios.

According to Table 1, as expected, in the Dependent Person scenario, all CNN models showed better results than the Independent Person one, since the network probably already knew samples similar to those used in the Dependent Person test dataset. These results are also consistent with those presented by Ye et al. [10], where the dependent person base also performed better than the independent person scenario.

The best result obtained was the VGG16 model, which had an average accuracy of 99.73% for the Dependent Person scenario and 70.21% for the Independent Person scenario. Among the models developed for mobile devices and embedded systems, the best result was the MobileNet model that had an average accuracy of 85.23% for the Dependent Person scenario and 61.35% for the Independent Person scenario.

The two most current CNN architectures used in this comparison, NASNetLarge and NASNetMobile, even with better results at ImageNet [18], had the worst results on the models tested. The NASNetLarge had an accuracy of 47.98% in the Dependent Person scenario and 9.8% in the Independent Person, whereas the NASNetMobile had an accuracy of 79.26% in the Dependent Person scenario, and 39.86% in the Independent Person Scenario.

We can also observe that the InceptionResNetV2 and VGG16 models also performed better than models designed for mobile devices and embedded systems in person dependent scenarios, which leads us to understand that they are learning better than the other architectures presented. However, considering the Independent Person scenario, the MobileNet model had an average accuracy of 61.35%, a value greater than the 48.35% achieved by InceptionResNetV2, signing that maybe this model is in overfitting, using the same architecture and parameters defined in its article that obtained the best accuracy in ImageNet Dataset [18].

The best result was that VGG16 model, which had an average accuracy of 99.73% for the Dependent Person scenario and 70.21% for the Independent Person scenario. Among the models developed for mobile devices and embedded systems, the best result was the MobileNet model that had an average accuracy of 85.23% for the Dependent Person scenario and 61.35% for the Independent Person scenario.

In Figure 03, we showed the accuracy curve during the training process for VGG16, which had the best results in this test, in the Independent Person scenario. According to Figure 3, we can note that around the 8th epoch, the training set was with an accuracy close to 100%. However, we followed the training for a few more epochs (about 25 epochs) until the accuracy of the validation set was stabilized (around 69%).

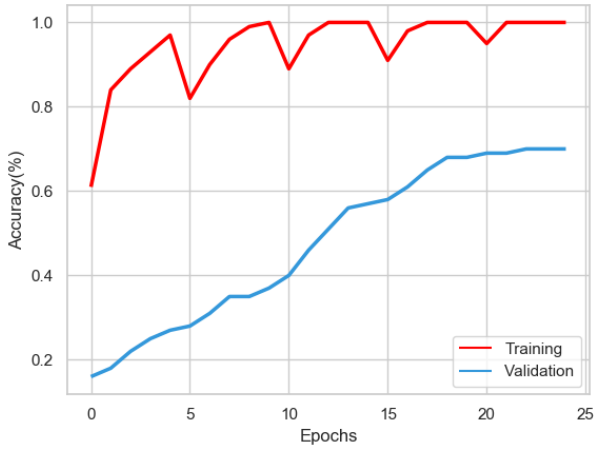


Figure 3: Increase in validation accuracy in independent person at each epoch.

Continuing the tests, we also generated a confusion matrix for the models with best results in the Independent Person scenario (VGG16 and MobileNet), to understand which were the possible causes of confusion, in which scenarios the models had difficulty for recognizing the letters, among others.. The results are shown in Figures 4 and 5.

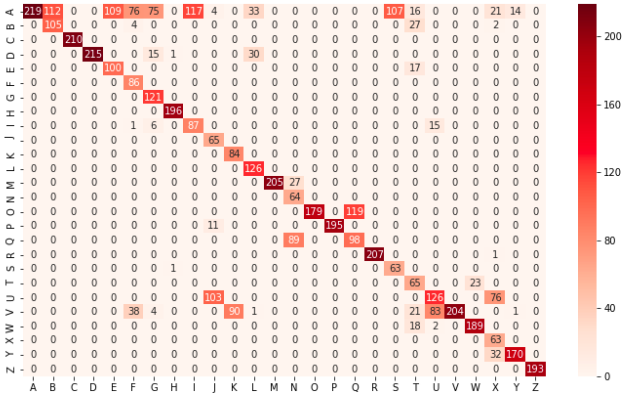


Figure 4: Confusion matrix for the VGG16 model in the Independent Person scenario.

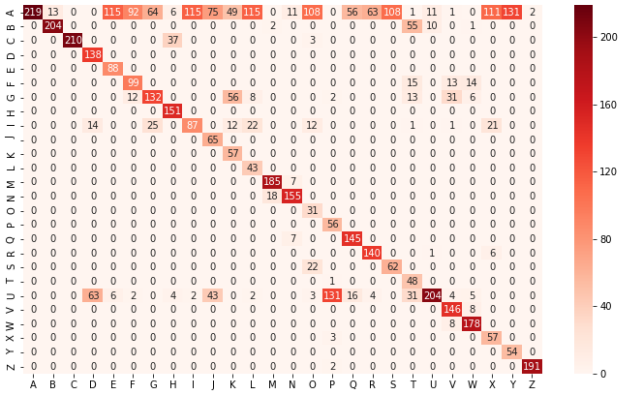


Figure 5: Confusion matrix for the MobileNet model in the Independent Person scenario.

According to Figures 4 and 5, we can observe that many false negatives are of letters that have similar signs, such as U and V (see Figure 6), N and Q (see Figure 7). Besides, the A letter, which has the largest number of false negatives, is similar to E, I and S signs (see Figure 8).



Figure 6: Image samples of the V and U signs, respectively, for the same person.

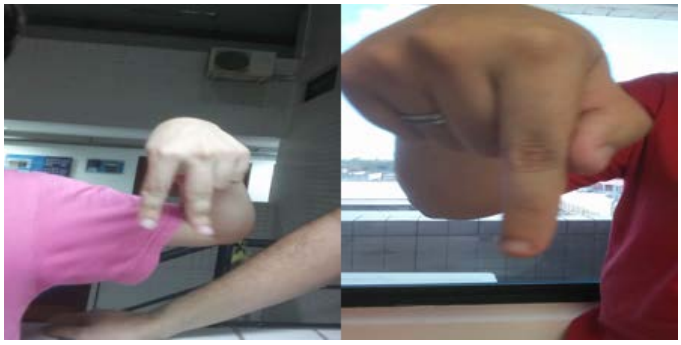


Figure 7: Image samples of the N and Q signs, respectively, for two different persons.



Figure 8: Image samples of the A, E, I and S signs, respectively, for four different volunteers.

Also, we also calculate the precision, recall, and F1-score values for the VGG16 and MobileNet models in the Independent Person scenario. The results are presented in Tables 2.

	MobileNet			VGG16		
Sign	**Precision**	**Recall**	**F1-Score**	**Precision**	**Recall**	**F1-Score**
A	0.15	1.00	0.26	0.24	1.00	0.39
B	0.75	0.94	0.83	0.76	0.48	0.59
C	0.84	1.00	0.91	1.00	1.00	1.00
D	1.00	0.64	0.78	0.82	1.00	0.90
E	1.00	0.42	0.59	0.85	0.48	0.61
F	0.70	0.48	0.57	1.00	0.42	0.59
G	0.51	0.60	0.55	1.00	0.55	0.71
H	1.00	0.76	0.87	1.00	0.99	0.99
I	0.45	0.43	0.44	0.80	0.43	0.56
J	1.00	0.36	0.52	1.00	0.36	0.52
K	1.00	0.33	0.49	1.00	0.48	0.65
L	1.00	0.23	0.37	1.00	0.66	0.80
M	0.96	0.90	0.93	0.88	1.00	0.94
N	0.90	0.86	0.88	1.00	0.36	0.52
O	1.00	0.17	0.30	0.60	1.00	0.75
P	1.00	0.29	0.45	0.95	1.00	0.97
Q	0.95	0.67	0.79	0.52	0.45	0.49
R	0.95	0.68	0.79	1.00	1.00	1.00
S	0.74	0.36	0.49	0.98	0.37	0.54
T	0.98	0.29	0.45	0.74	0.40	0.52
U	0.39	0.90	0.55	0.41	0.56	0.47
V	0.95	0.72	0.82	0.46	1.00	0.63
W	0.96	0.84	0.89	0.90	0.89	0.90
X	0.95	0.29	0.45	1.00	0.32	0.49
Y	1.00	0.29	0.45	0.84	0.92	0.88
Z	0.99	0.99	0.99	1.00	1.00	1.00
Total	0.84	0.61	0.64	0.83	0.70	0.71

Table 2: Precision, recall and f1-score values for the MobileNet and VGG16 models in the Independent Person scenario.

According to Tables 2, the average values of precision, recall, and F1-score for the VGG16 model were 0.83, 0.70, and 0.71, respectively. For the MobileNet model, the average values for the precision, recall and f1-score were 0.84, 0.61 and 0.64, respectively. We can note that these values, especially the f1-score value, are similar to the average accuracy obtained by these models (see Table 1).

From these results, we can also observe that models had better performance when the network knew the hand and bodies of the volunteer (Dependent Person Scenario). To solve this problem, a possible solution would include the possibility of the user performing the calibration of the model, making the capture of a set of images of his hand and sending it to a remote service, with proper hardware, to make a "custom training". Thus, this calibration would only be necessary in the first use, which bring few difficulties for the application of the solution in a real-world scenario. Developing and testing this calibration function is one of our proposals for future work.

To improve our results, especially in the dependent person scenario, the most intuitive solution is to expand the base to have more variations in its instances out of artificial techniques like data augmentation, what would it be from new videos of volunteers and scenarios in the dataset, even signals beyond those used in fingerspelling, which can be useful in future works, that would come with the reception of dataset in the scientific community. Knowing this is a slow process, we also plan to improve our preprocessing steps and investigate another models and architectures.

The preprocessing approach, use the raised base in conjunction with an image recognition technique, who could be another convolutional neural network, aiming first to identify in the images the location of the hands, arms, chest, face and from this, crop the image, leaving it with only the important parts in signal recognition.

The investigation of another models and architectures would be based in improvements in the models which have the bests results in our tests, such as VGG16 and MobileNet.

5 FINAL REMARKS

In this work, we presented a solution proposal for fingerspelling recognition in Brazilian Sign Language using Convolutional Neural Networks. The solution does not use any gloves, armbands or visual markers in the images, i.e. it recognizes the Libras gestures directly from the images, which makes its use more easily applicable in practical scenarios involving machine translation from sign languages into oral languages (eg., from Libras to Brazilian Portuguese).

To develop the solution, we create a dataset containing approximately 224000 images, which represents 26 letters of the Libras alphabet signed by 12 persons in different backgrounds, body arm and hand positions, and different lighting patterns.

We also performed a set of tests with four CNN models, considering two scenarios: Dependent Person, and Independent Person. The results show that the best models had an accuracy of approximately 99% in the Dependent Person scenario, and approximately 71% in the Independent Person scenario. Since the Independent Person scenario have a worse result and the Dependent Person had a very good result, one possible solution to improve the solution would the user perform a calibration of the model, making the capture of his own hand, and retraining the model. The development of this approach is one proposal for future work. Another proposal for future work is to investigate the use of CNN3D, for recognizing Libras signs in videos.

ACKNOWLEDGMENTS

We gratefully acknowledge the support of NVIDIA Corporation with the donation of the Quadro P6000 GPU used for this research.

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